

Rising Waters

Flood Management System — AI/ML Project

Solution Requirements

Phase 2: Requirement Analysis

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

Functional Requirements

FR-01	System shall accept 6 meteorological inputs via a web form
FR-02	System shall validate all inputs before processing
FR-03	System shall scale inputs using pre-fitted StandardScaler
FR-04	System shall load XGBoost model and return binary classification
FR-05	System shall display dedicated Flood Chance result page (red)
FR-06	System shall display dedicated No Flood Chance result page (green)
FR-07	System shall log every prediction to predictions_log.csv
FR-08	System shall display all past predictions on History page
FR-09	System shall show model accuracy comparison on Dashboard

Non-Functional Requirements

NFR-01 Performance	Prediction response time < 2 seconds
NFR-02 Accuracy	XGBoost model test accuracy >= 95%
NFR-03 Availability	System available 24/7 via IBM Cloud
NFR-04 Usability	Responsive design, accessible on mobile and desktop
NFR-05 Security	Session-based result passing, no raw model exposure
NFR-06 Scalability	Dockerized deployment, horizontal scaling ready
NFR-07 Maintainability	Modular code: separate utils, model, templates directories